UNIVERSITY OF CAMBRIDGE

MPhil in Data Intensive Science

Lent Term 2026

S2 Major Module Coursework
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Attempt all parts of the coursework.

The anticipated number of marks allocated to each part of a question is indicated in the right margin, to be assessed from the code you provide and your written answers.

This coursework should be submitted via a GitLab repository which will be created for you. Place all of your code and your written answers into this repository. Your written answers should be in a file called `written_answers.pdf` *along with any figures. You will be provided access to your repository until the deadline of 23:59, April 1st 2026, after which your work will be deemed ready for assessment.*

Your written answers should be brief and should not exceed the specified number of words when this is indicated in the question. Any plots asked for in the question should also be included along with your written answers (these are not included in the word count).

You are expected to submit code and associated material that demonstrates good software development practices as covered in the Research Computing module.

You are reminded to comply with the requirements given in the Course Handbook regarding the use of, and declaration of use of, autogeneration tools. This declaration should be included in your written answers (this is not included in the word count).

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DRAFT 1

This coursework asks you to use and compare different methods of calculating model evidences, Bayes factors and odds ratios for Bayesian model comparison.

In this coursework, you will analyse the rate of serious mining accidents in the UK from 1851-1962 (a period of $L = 40550$ days) during which there were a total of $N = 191$ explosions in which 10 or more people were killed. The intervals I_i (in days) between accidents are provided in the data file `coal_mining_accident_data.dat` released with the coursework;

$$\{I_i | i = 1, 2, \ldots, N-1\} = (157, 123, 2, 124, \ldots, 1358, 2366, 952, 632).$$

This data is from Jarrett (1979). On 06/12/1875 there were two accidents with an interval of 0 days. Already, just from this data snippet, you might conjecture that the accident rate decreased over time. You will investigate this using a multiple change-point analysis.

This coursework follows the notation and conventions of Green (1995) "*Reversible jump Markov chain Monte Carlo computation and Bayesian model determination*" Biometrika, Vol. 82, No. 4, 711-732. `https://doi.org/10.1093/biomet/82.4.711`. Note that the accident data set analysed by Green (1995) is NOT exactly the same as that used here.

The model M_k for the rate of accidents uses a inhomogeneous Poisson process with a rate x that is constant except at k steps, or change points, at positions $0 < s_1 < s_2 < \ldots < s_k < L$ where the rate changes discontinuously. The rate of accidents for times $s_j < t < s_{j+1}$ is h_j, for $j = 0, 1, \ldots, k$. Therefore, the model M_k has $2k + 1$ parameters: the k change points s_j and the $k + 1$ rates, also known as the heights, h_j. In model M_k the piecewise-constant rate of accidents as a function of time t is

$$x(t) = \begin{cases} h_0 & \text{if } 0 \le t < s_1 \\ h_1 & \text{if } s_1 \le t < s_2 \\ & \ldots \\ h_k & \text{if } s_k \le t < L \end{cases}.$$

Let y_i denote the time of the ith accident. The log-likelihood function is

$$\log \mathcal{L}(\{I_i\} | \{s_j\}, \{h_j\}, M_k) = \sum_{i=1}^{N} \log x(y_i) - \int_0^L \mathrm{d}t \; x(t)$$

The prior probabilities for the models M_k are specified by taking k to be distributed as a Poisson random variable with mean parameter $\lambda = 3$ subject to the condition that $k \le 30$;

$$\pi(k) \propto \frac{\lambda^k \exp(-\lambda)}{k!}, \quad \text{for} \quad k = 0, 1, \ldots, 30.$$

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1 **Accident data.** Load the intervals I_i from the `coal_mining_accident_data.dat` file. Plot the cumulative number of accidents over time. What is the mean rate of accidents over the total period covered by the data? [2]

2 **Priors.** Model M_k has $2k+1$ parameters: the k change point positions s_j and the $k+1$ heights h_j . *A priori*, the heights are taken to be identically and independently distributed as $h_j \sim \Gamma(\alpha, \beta)$ with $\alpha = 1$ and $\beta = 200$ days and the positions $s_1, s_2, \ldots, s_k$ are taken to be distributed as the even-numbered order statistics from $2k + 1$ points independently uniformly distributed on the interval $[0, L]$.

[*A gamma-distributed random variable $h \sim \Gamma(\alpha, \beta)$ with shape and rate parameters α and β has PDF $P(h) = \beta^\alpha \exp(-\beta h)/\Gamma(\alpha)$, for $h > 0$.*

The even-numbered order statistics are distributed as follows. Consider standard uniform random variates $u_i \sim U(0, 1)$ independent for $i = 1, 2, \ldots, k + 1$. The order statistics $u_{(j)}$ are defined by sorting these into ascending order: $0 < u_{(1)} < u_{(2)} < \ldots < u_{2k+1} < 1$. The even-numbered order statistics are $\hat{s}_j = u_{2j}$, for $j = 1, 2, \ldots, k$. The change points are then scaled to lie in the desired interval; $s_j = L\hat{s}_j$]

(a) An alternative prior for the change point positions would be to just take the k plain order statistics. This would involve drawing k independent uniform random variables on $[0, L]$ and sorting into ascending order. Describe qualitatively (in under 200 words) the difference between this prior and the even-numbered order statistics prior. Illustrate your answer using a suitable plot of the two priors for $k = 4$. [4]

(b) Hereafter, use the even-numbered order statistics prior for the change points s_j described above. Show that the even-numbered order statistics prior on the change point positions has probability density function (PDF) [6]

$$\pi(s_1, s_2, \ldots, s_k | M_k) \propto \prod_{j=1}^{k+1} g_j,$$

where the gaps g_j between change points are

$$g_1 = s_1, \quad g_2 = s_2 - s_1, \quad \ldots \quad g_k = s_k - s_{k-1}, \quad g_{k+1} = L - s_k.$$

[*You may use without proof the fact that the PDF of the plain order statistics is a constant, $P(u_{(1)}, u_{(2)}, \ldots, u_{(2k+1)}) = (2k + 1)!$ on the region where the distribution has support, $0 < u_{(1)} < u_{(2)} < \ldots < u_{(2k+1)} < 1$.*]

This can be recognised as being the PDF of a Dirichlet distribution with shape parameters $\boldsymbol{\alpha} = (2, 2, \ldots, 2) \in \mathbb{R}^{k+1}$. This allows us to find the normalisation of the prior on the change point positions;

$$\pi(s_1, s_2, \ldots, s_k | M_k) = \frac{1}{L^{2k+1} B(\boldsymbol{\alpha})} \prod_{j=1}^{k+1} g_j.$$

where $B(\boldsymbol{\alpha})$ is the multivariate beta function.

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3 **The constant rate model, M_0.** This model has a single parameter, h_0.

(a) Make a plot comparing the prior $\pi(h_0|M_0)$ and posterior $P(h_0|\{I_i\}, M_0)$ for the model M_0. Also calculate the evidence $Z_0 \equiv P(\{I_i\}|M_0)$ for this model by a direct one-dimensional numerical integration. [6]

4 **The 1 change point model, M_1.** This model has three parameters: s_1, h_0, and h_1.

(a) Use an MCMC package of your choice to sample the posterior for model M_1. Make a corner plot for this posterior. Briefly explain (in under 300 words) which algorithm/package you used for the MCMC, any important settings for this algorithm, and any diagnostic checks you performed to check for convergence. [8]

(b) Explain briefly (in under 200 words) how the Savage Dickey density ratio can, in principle, be used to find the ratio of evidences between the models M_1 and M_0. Also explain what difficulty is encountered when trying to use the Savage Dickey density ratio in this example. You should illustrate your explanation with a plot. [3]

(c) Explain briefly (in under 200 words) what difficulty is encountered if one tried to use the Laplace approximation to estimate the evidence $Z_1 \equiv P(\{I_i\}|M_1)$ for the model M_1. [3]

(d) Use a nested sampling algorithm of your choice to evaluate the evidence for the model M_1. Using your answer from 3(a), give both the evidence ratio and posterior odds ratio for the pair of models M_0 and M_1. Describe briefly (in under 200 words) the details of the algorithm and any relevant settings used. [8]

5 **The k change point models, M_k.** The full space of all the models $\{M_k|k = 0, 1, \ldots, 30\}$ can be explored simultaneously using a reversible jump MCMC. The file `RJMCMC.py` released with the coursework contains a partially written class that implements the algorithm described in Green (1995); you should read that paper, and section 4.3 in particular, for full details of the algorithm.

(a) Having read and understood the contents of `RJMCMC.py`, you should complete the class by implementing the method `height_change_move` and the expressions for the log prior ratio in both the `birth_move` and `death_move` methods as described in section 4.3 of Green (1995). Areas where you are required to edit are marked with a `# TO DO:` comment in the code. [10]

(b) Run the reversible jump MCMC sampler for at least 10^5 iterations. (If you feel like giving your laptop a work out, you could run a bit longer, but there is no need to go way beyond this.) Plot the trace plot for the model index k to verify that the chain has fully explored the model space. Plot the posterior on the model index k. What is the preferred maximum a posteriori number of change points? [6]

(c) Plot the inferred rate $x(t)$ as a function of time from your inference marginalised over the number of change points (i.e. model averaging). Your plot should include shaded regions indicating the 50% and 90% uncertainties on the inferred rate.

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Briefly explain (in under 250 words) how you calculated the uncertainty regions. [4]

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